AP PRECALCULUS · TOPIC 1.10

Rational Functions and Holes

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $r(x) = \dfrac{x^2 + 2x - 8}{x^2 - 3x + 2}$. What are the coordinates of the hole in the graph of r?

A. $(2, 6)$
B. $(2, 0)$
C. $(-2, 6)$
D. $\left(2, \dfrac{1}{6}\right)$

2 Which statement about the graph of $r(x) = \dfrac{x^2 - 9}{x^2 - 6x + 9}$ is true?

A. A hole at $x = 3$ and no vertical asymptote
B. A vertical asymptote at $x = 3$ and no hole
C. A hole at $x = 3$ and a vertical asymptote at $x = -3$
D. A hole at $x = -3$ and a vertical asymptote at $x = 3$

3 What is the value of $\lim\limits_{x \to -3} \dfrac{x^2 + 2x - 3}{x + 3}$?

A. 0
B. 2
C. -4
D. The limit does not exist

4 For a rational function r, the outputs for inputs near 2 are shown. Which statement is supported by the table?

x	1.9	1.99	1.999	2.001	2.01	2.1
$r(x)$	4.9	4.99	4.999	5.001	5.01	5.1

A. The graph of r has a vertical asymptote at $x = 2$
B. The graph of r has a hole at $(5, 2)$
C. The graph of r has a zero at $x = 2$
D. The graph of r has a hole at $(2, 5)$

5 Which function has a hole at $x = 1$ and a vertical asymptote at $x = -2$?

A. $r(x) = \dfrac{(x-1)(x+3)}{(x-1)(x+2)}$
B. $r(x) = \dfrac{(x+1)(x+3)}{(x+1)(x-2)}$
C. $r(x) = \dfrac{(x+2)(x+3)}{(x-1)(x+2)}$
D. $r(x) = \dfrac{x+3}{(x-1)(x+2)}$

6 Let $r(x) = \dfrac{(x-2)^3(x+1)}{(x-2)^2(x-5)}$. What are the coordinates of the hole in the graph of r?

A. $(2, 3)$
B. $(2, 0)$
C. $(-1, 0)$
D. $(5, 0)$

7 How many holes and how many vertical asymptotes does the graph of $r(x) = \dfrac{(x+1)(x-2)(x-4)}{(x-2)(x-4)(x+5)}$ have?

A. 0 holes and 3 vertical asymptotes
B. 1 hole and 2 vertical asymptotes
C. 2 holes and 1 vertical asymptote
D. 3 holes and 0 vertical asymptotes

8 What are the coordinates of the hole in the graph of $r(x) = \dfrac{2x^2 - 8}{x^2 - x - 2}$?

A. $(2, 2)$
B. $\left(2, \dfrac{4}{3}\right)$
C. $\left(2, \dfrac{3}{8}\right)$
D. $\left(2, \dfrac{8}{3}\right)$

AP PRECALCULUS · TOPIC 1.10 · SECTION 2

Rational Functions and Holes

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A sprinter's distance from the start, in metres, is $s(t) = t^2 + 3t$ for t seconds after the start. The average speed between 1 second and t seconds is $v(t) = \dfrac{s(t) - s(1)}{t - 1} = \dfrac{t^2 + 3t - 4}{t - 1}$, for $t > 0$ and $t \neq 1$.

(a) Factor the numerator, write $v(t)$ in simplified form and give the coordinates of the hole in the graph of v. DOK3 2 pt

(b) Use limit notation to describe the behaviour of $v(t)$ near $t = 1$, and interpret the result in context. DOK3 2 pt

(c) Find the time at which the average speed is 7.5 metres per second. Explain why no time gives an average speed of exactly 5 metres per second. DOK3 2 pt

Question 10 No calculator · 6 points

Let $r(x) = \dfrac{(x-3)^2(x+1)(x-2)}{(x-3)(x-2)^2(x+4)}$.

(a) For each of the inputs $x = 3$ and $x = 2$, compare the multiplicity of the zero in the numerator with its multiplicity in the denominator, and state whether the graph has a hole or not. DOK3 2 pt

(b) Find the coordinates of the hole. DOK3 2 pt

(c) State the equations of all vertical asymptotes and the x-intercepts of the graph of r. DOK3 2 pt